

Physics for Engineers 2 Lab

KIRCHHOFF'S LAWS

MPS Department | FEU Institute of Technology



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OBJECTIVES

- To verify the Kirchhoff's voltage law
- To verify the Kirchhoff's current law



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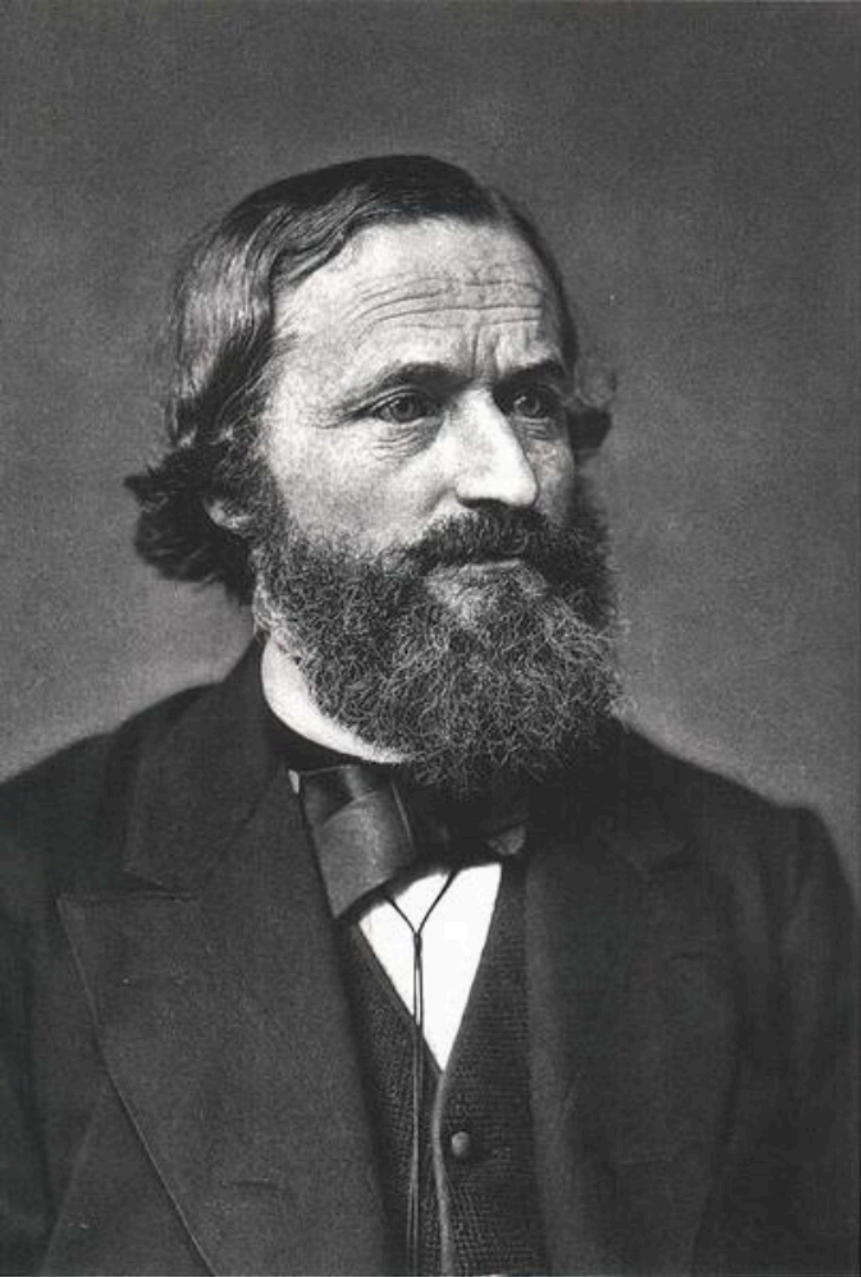


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Gustav Robert Kirchhoff(1824-1887)



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KIRCHHOFF'S LAWS

Kirchhoff's laws are a set of laws that quantify how current flows through a circuit and how voltage varies around a loop in a circuit. They are used to govern the conservation of charge and energy in standard electrical circuits. Two significant circuital laws are applied in every simple and complex electrical circuit in physics. These laws were postulated in 1845 by German physicist Gustav Kirchhoff.

Figure 1. *Gustav Robert Kirchhoff*. Retrieved from <https://byjus.com/physics/kirchhoffs-law/>

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KIRCHHOFF'S CURRENT LAW

Kirchhoff's first law is known as Kirchhoff's Current Law (KCL). It is also known as Kirchhoff's junction rule, Kirchhoff's nodal rule, and Kirchhoff's point rule.

The law states that at any node, the sum of Current entering and leaving is zero. This is due to conservation of charges.

$$\sum I_{entering} + \sum I_{leaving} = 0$$

A node is an intersection of branches in a circuit.

For the sign convention: $I_{entering}$ (+)

$I_{leaving}$ (-)



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KIRCHHOFF'S VOLTAGE LAW

Kirchhoff's second law is known as Kirchhoff's Voltage Law (KVL).

The law states that at any loop, the sum of voltages is zero. This is due to conservation of energy.

$$\sum v = 0$$

A loop is any closed path.

KIRCHHOFF'S VOLTAGE LAW

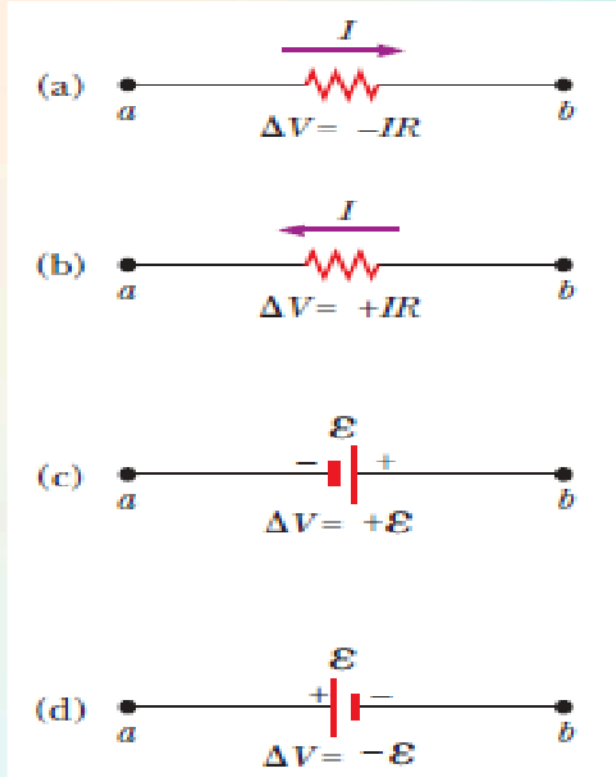


Figure 1. Rules for determining the potential differences across a resistor and a battery, by Serway, *Physics for Scientists and Engineers*, 7th Ed. p. 786

(a) Charges move from the high-potential end of a resistor toward the low potential end, so if a resistor is traversed in the direction of the current, the potential difference V across the resistor is **$-IR$** .

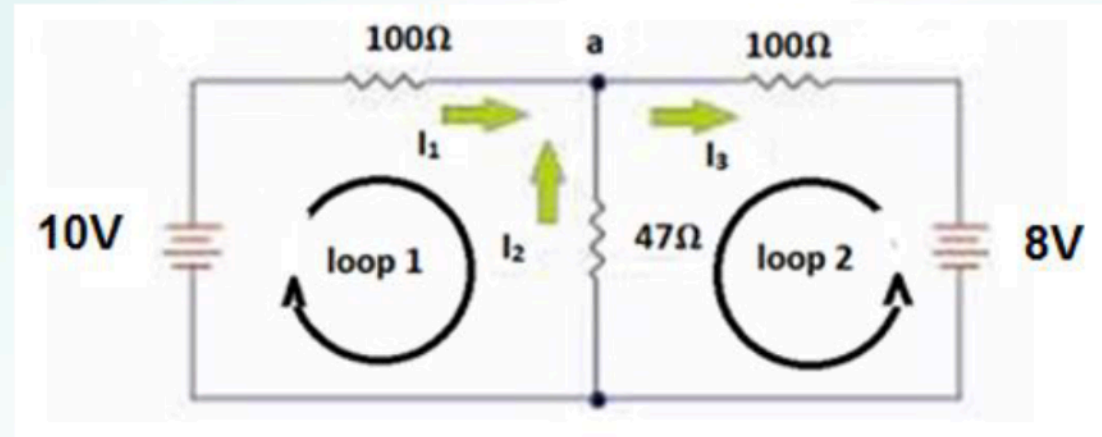
(b) If a resistor is traversed in the direction opposite the current, the potential difference V across the resistor is **$+IR$** .

(c) If a source of emf (assumed to have zero internal resistance) is traversed in the direction of the emf (from negative to positive), the potential difference V is **$+\mathcal{E}$** .

(d) If a source of emf (assumed to have zero internal resistance) is traversed in the direction opposite the emf (from positive to negative), the potential difference V is **$-\mathcal{E}$** .

KIRCHHOFF'S LAW

This is the circuit used in the experiment.



Data and Results

Use the following table and templates for the data and results

Table 8.1 Kirchhoff's Current Law Verification		
Current	Measured	Calculated
I_1	+	+
I_2	+	+
I_3	-	-
Sum		

Table 8.2 Kirchhoff's Voltage Law Verification					
Loop 1			Loop 2		
	Measured	Calculated		Measured	Calculated
V_1	-	-	V_2	-	-
V_2	-	-	V_3	-	-
$\mathcal{E}_1$	+	12V	$\mathcal{E}_2$	+	8V
Sum			Sum		

REFERENCES

Serway, R. and Jewett, J. (2008). Rules for determining the potential differences across a resistor and a battery. [Photograph] *Physics for Scientists and Engineers with Modern Physics*, 7th Ed. Thompson Learning Inc., Belmont, CA

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